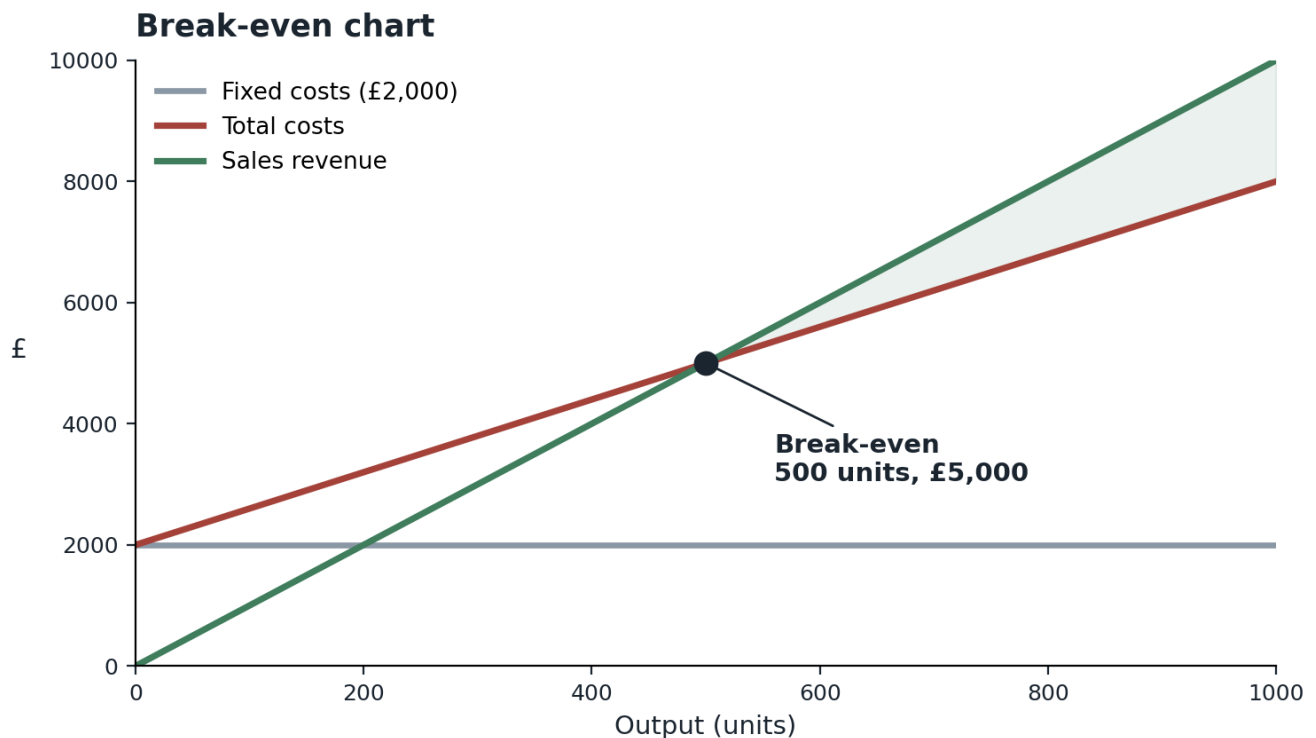


Break-Even Made Visual

The chart, the formula, a worked example and one to try



The formula

Break-even (units) =

Fixed costs ÷ contribution per unit

where contribution per unit = selling price – variable cost per unit

Worked example

Fixed costs £2,000 · price £10 · variable cost £6.

Contribution = £10 – £6 = **£4**

Break-even = 2,000 ÷ 4 = 500 units (= £5,000 of sales)

Now you try

A firm has fixed costs of £6,000. It sells each unit for £15; the variable cost is £9 per unit. What output does it need to break even?

Answer: contribution = £15 – £9 = £6 → 6,000 ÷ 6 = 1,000 units.

What the chart shows

After the break-even point, the gap between revenue and total cost is **profit** (shaded). Before it, the firm makes a loss. The **margin of safety** is how far current output sits above break-even. If fixed costs rise, the total-cost line shifts up and the break-even output increases.

Margin of safety = current output - break-even. If this firm makes 800 units, its margin of safety is 800 - 500 = 300 units - how far sales can fall before it makes a loss.

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